Q1. What is the correct file extension for Python files?

A. .pt

B. .py [Correct]

C. .pyt

D. .python

Explanation: Python scripts and modules use the '.py' extension so they can be recognized and executed by the Python interpreter.

Q2. Which data structure is immutable in Python?

A. List

B. Set

C. Dictionary

D. Tuple [Correct]

Explanation: Tuples are immutable, meaning their elements cannot be changed, added, or removed after creation, unlike Lists, Sets, and Dictionaries.

Q3. What is the output of: len([1, 2, 3, 4])?

A. 3

B. 4 [Correct]

C. 5

D. Error

Explanation: The len() function returns the total number of items in an iterable. The list [1, 2, 3, 4] contains exactly 4 integer elements.

Q4. What is Pandas mainly used for?

A. Machine learning

B. Numerical computation

C. Data manipulation and analysis [Correct]

D. 3D graphics

Explanation: Pandas is a powerful library built on NumPy specifically designed for structured data manipulation, analysis, and cleaning using DataFrames and Series.

Q5. What does df.head() do in Pandas?

A. Returns last 5 rows

B. Returns first 5 rows [Correct]

Explanation: By default, the .head() method returns the first 5 rows of a DataFrame to help quickly inspect the data.

Q6. Which method is used to read a CSV file using Pandas?

A. read_file

B. read_csv [Correct]

C. open_csv

D. load_csv

Explanation: The pd.read_csv() function is the standard method in Pandas for loading comma-separated values (CSV) into a DataFrame.

Q7. What library is commonly used for numerical computing in Python?

A. Pandas

B. NumPy [Correct]

C. TensorFlow

D. Seaborn

Explanation: NumPy provides support for large, multi-dimensional arrays and matrices, along with mathematical functions to operate on these arrays efficiently.

Q8. What does the .shape attribute of a DataFrame return?

A. Size of a column

B. Number of columns

C. Tuple of (rows, columns) [Correct]

D. Data types

Explanation: The .shape attribute returns a tuple representing the dimensionality of the DataFrame, formatted as (number_of_rows, number_of_columns).

Q9. Which of the following is not a data type in Python?

A. Integer

B. Float

C. Character [Correct]

D. String

Explanation: Python does not have a distinct "character" data type. A single character is simply treated as a string of length 1.

Q10. How do you insert comments in Python code?

A. //

B. --

C. # [Correct]

D. %%

Explanation: The hash symbol (#) is used in Python to indicate a single-line comment, which is ignored by the interpreter.

Q11. What is the keyword to define a function in Python?

A. def [Correct]

B. func

C. lambda

D. function

Explanation: The 'def' keyword is used to declare and define a new custom function block in Python.

Q12. What is the correct way to import the NumPy library?

A. import numpy as np [Correct]

B. import NumPy

C. include numpy

D. use numpy

Explanation: The standard convention is to import numpy with the alias 'np' for shorter syntax when calling its methods.

Q13. How do you handle exceptions in Python?

A. if-else

B. do-while

C. try-except [Correct]

D. for-else

Explanation: Python uses the try-except block to catch and handle runtime errors gracefully without crashing the program.

Q15. Which function in NumPy returns an array of zeros?

A. np.ones()

B. np.zeros() [Correct]

C. np.empty()

D. np.null()

Explanation: np.zeros(shape) creates a new array of the given shape and data type, filled entirely with zeros.

Q16. What is PySpark used for?

A. Image processing

B. Web development

C. Big data analytics [Correct]

D. UI design

Explanation: PySpark is the Python API for Apache Spark, a distributed computing framework widely used for processing and analyzing massive datasets.

Q17. Which of the following is not a big data characteristic?

A. Volume

B. Variety

C. Velocity

D. Visibility [Correct]

Explanation: The primary 'V's of big data are Volume, Velocity, Variety (and often Veracity). Visibility is not considered a core defining characteristic.

Q18. What is the default delimiter in read_csv()?

A. Tab

B. Space

C. Comma [Correct]

D. Semicolon

Explanation: 'CSV' stands for Comma-Separated Values, so the default delimiter (separator) used by read_csv is naturally the comma.

Q20. What is the correct syntax for a lambda function?

A. lambda x: x + 2 [Correct]

B. def x: return x + 2

C. func x => x + 2

D. lambda(x) = x + 2

Explanation: A lambda function is an anonymous inline function defined using the syntax `lambda arguments: expression`.

Q21. How do you check missing values in a Pandas DataFrame?

A. df.is_null()

B. df.hasnull()

C. df.isna() [Correct]

D. df.missing()

Explanation: df.isna() (or df.isnull()) returns a boolean same-sized object indicating whether the values are NA (missing).

Q22. What does the map() function do in Python?

A. Maps keys to values

B. Applies a function to a list [Correct]

C. Joins two lists

D. Reduces a list

Explanation: The map() function applies a specified function to every item of an iterable (like a list) and returns an iterator of the results.

Q23. Which function writes a DataFrame to a CSV?

A. to_csv [Correct]

B. export_csv

C. save_csv

D. write_csv

Explanation: The DataFrame.to_csv() method writes the DataFrame's contents into a comma-separated values file.

Q25. What Python library is used to visualize data?

A. os

B. pandas

C. matplotlib [Correct]

D. sklearn

Explanation: Matplotlib is the foundational plotting and visualization library in Python, used for creating static, animated, and interactive graphs.

Q26. What is the purpose of .dropna() in Pandas?

A. Replaces NA with zero

B. Removes columns

C. Drops rows with NA values [Correct]

D. Adds NA values

Explanation: The .dropna() method is used to remove rows (or columns, depending on axis) that contain missing or NA values.

Q27. What type of loop does Python support?

A. for [Correct]

B. foreach

C. loop

D. iterate

Explanation: Python primarily supports 'for' and 'while' loops. It does not have 'foreach', though its 'for' loop acts identically to a foreach loop over iterables.

Q28. Which function returns the maximum value in a list?

A. sum()

B. max() [Correct]

C. top()

D. high()

Explanation: The max() function computes and returns the largest item from an iterable or from two or more arguments.

Q30. Which of these is a Python big data framework?

A. PyTorch

B. Scikit-learn

C. PySpark [Correct]

D. Matplotlib

Explanation: PySpark interfaces with Apache Spark, providing distributed processing capabilities necessary for handling big data.

Q31. What is an ETL pipeline?

A. A database query

B. Extract, Transform, Load [Correct]

C. Email transfer log

D. External tracking list

Explanation: ETL stands for Extract (getting data), Transform (cleaning/formatting), and Load (storing it into a data warehouse or database).

Q32. Which file format is commonly used with big data?

A. .docx

B. .parquet [Correct]

C. .exe

D. .mp4

Explanation: Parquet is a highly efficient, columnar storage file format optimized for fast data processing and querying in big data ecosystems like Spark.

Q33. What does the os module allow in Python?

A. Network communication

B. File and directory access [Correct]

C. UI design

D. Image processing

Explanation: The 'os' module provides a portable way of using operating system dependent functionality like reading/writing directories and handling paths.

Q35. What's the correct syntax for opening a file for reading?

A. open("file.txt", "w")

B. open("file.txt", "r") [Correct]

C. read("file.txt")

D. file.read("file.txt")

Explanation: The built-in open() function takes the filename and the mode. "r" denotes read mode, which is also the default if omitted.

Q36. What is the output of: 3 == 3.0?

A. True [Correct]

B. False

C. Error

D. None

Explanation: Python evaluates the numeric values for equality. Since the integer 3 and the float 3.0 hold the same value, the expression evaluates to True.

Q37. What is SparkSession in PySpark?

A. GUI

B. Starting point for DataFrame APIs [Correct]

C. Logging system

D. File manager

Explanation: Introduced in Spark 2.0, SparkSession provides a single point of entry to interact with underlying Spark functionality and allows programming with DataFrames.

Q38. Which of the following is a transformation in PySpark?

A. collect()

B. show()

C. map() [Correct]

D. count()

Explanation: map() is a transformation (returns a new RDD/DataFrame lazily), whereas collect(), show(), and count() are actions that trigger computation.

Q40. Which library supports parallel computing with arrays?

A. pandas

B. NumPy

C. Dask [Correct]

D. matplotlib

Explanation: Dask natively scales NumPy, Pandas, and Scikit-Learn workflows to utilize parallel, distributed computing across clusters.

Q41. What keyword is used for class definition?

A. object

B. define

C. class [Correct]

D. struct

Explanation: The 'class' keyword is used to create a new user-defined blueprint (class) for creating objects in Object-Oriented Python.

Q42. Which method is used to get summary statistics in Pandas?

A. df.summary()

B. df.describe() [Correct]

C. df.info()

D. df.stats()

Explanation: df.describe() computes various summary statistics (mean, std, min, max, quartiles) for numeric columns in a DataFrame.

Q43. How do you import all functions from a module?

A. import module

B. import module.*

C. from module import * [Correct]

D. using module

Explanation: The asterisk (*) acts as a wildcard, importing all public functions and variables from the specified module into the current namespace.

Q45. What is an RDD in PySpark?

A. Raw Data Driver

B. Reliable Data Description

C. Resilient Distributed Dataset [Correct]

D. Remote Data Directory

Explanation: Resilient Distributed Dataset (RDD) is the fundamental, fault-tolerant data structure of Apache Spark that represents distributed collections of objects.

Q46. Which library provides machine learning in Spark?

A. SparkML [Correct]

B. SparkViz

C. SparkAI

D. SparkStats

Explanation: Spark MLlib (or SparkML for DataFrame API) is Apache Spark's scalable machine learning library.

Q47. Which module helps with JSON in Python?

A. os

B. pandas

C. json [Correct]

D. text

Explanation: The built-in 'json' module provides methods like load(), loads(), dump(), and dumps() for encoding and decoding JSON data.

Q48. What does json.loads() do?

A. Reads file

B. Converts string to JSON [Correct]

C. Converts JSON to string

D. Saves JSON

Explanation: json.loads() parses a JSON-formatted string and converts it into a corresponding Python dictionary object.

Q50. What does df['col'].value_counts() do?

A. Counts unique values [Correct]

B. Shows null values

C. Sums column

D. Sorts values

Explanation: value_counts() returns a Series containing counts of unique values in descending order, useful for finding frequencies.

Q51. What is the main function of SparkContext in PySpark?

A. Create charts

B. Connect to MySQL

C. Interface with Spark cluster [Correct]

D. Define DataFrames

Explanation: SparkContext is the core entry point for low-level Spark functionality, allowing the application to connect to the Spark cluster and coordinate tasks.

Q52. What is the purpose of .cache() in PySpark?

A. Delete cache

B. Optimize DataFrame performance [Correct]

C. Remove duplicates

D. Sort dataset

Explanation: .cache() stores the DataFrame or RDD in memory, significantly speeding up subsequent actions that reuse this dataset.

Q53. In PySpark, what does .show() do?

A. Displays the schema

B. Prints the first 20 rows [Correct]

Explanation: By default, the .show() action evaluates the DataFrame and prints its first 20 rows to the console

in a tabular format.

Q54. Which Python library is used for working with structured tabular data?

A. NumPy

B. TensorFlow

C. Pandas [Correct]

D. Matplotlib

Explanation: Pandas provides DataFrames and Series, which are the standard structures in Python for working with labeled or tabular data.

Q55. What type of data format is Apache Avro?

A. Binary JSON-based [Correct]

B. Text-based XML

C. CSV

D. HTML

Explanation: Apache Avro is a data serialization system that relies on JSON for defining data types and protocols, but serializes data into a compact binary format.

Q56. What is the purpose of with open('file.txt') as f:?

A. Opens file without closing

B. Opens file and auto-closes [Correct]

C. Writes to file

D. Deletes the file

Explanation: Using 'with' creates a context manager that ensures the file is safely and automatically closed after the block completes, even if an exception occurs.

Q57. What does SparkSession.builder do?

A. Displays logs

B. Creates a SparkSession [Correct]

C. Reads JSON

D. Deletes session

Explanation: SparkSession.builder provides the builder pattern methods to configure and instantiate a

SparkSession object.

Q59. How do you filter rows in a Pandas DataFrame?

A. df.select()

B. df.filter_rows()

C. df[df['column'] > 5] [Correct]

D. df.filterBy()

Explanation: Boolean indexing using square brackets allows filtering rows where the condition (e.g., column > 5) is True.

Q60. Which library is used to connect Python with Kafka?

A. confluent_kafka [Correct]

B. streamlit

C. dask_kafka

D. sparkstream

Explanation: confluent_kafka is a high-performance Python client for Apache Kafka, widely used for building streaming data pipelines.

Q61. What?s the difference between a list and a tuple in Python?

A. Tuple is mutable

B. List is immutable

C. Tuple is immutable [Correct]

D. Both are the same

Explanation: Lists can be altered after creation (mutable), while tuples cannot be changed, appended, or modified once instantiated (immutable).

Q62. What function is used to apply an operation to each row in Pandas?

A. apply() [Correct]

B. forEach()

C. each()

D. transform()

Explanation: The apply() method invokes a function along an axis of the DataFrame (either rows with axis=1

or columns with axis=0).

Q64. Which of these is a columnar storage format optimized for Spark?

A. JSON

B. Parquet [Correct]

C. TXT

D. XML

Explanation: Parquet organizes data by column instead of row, allowing Spark to read only required columns, which minimizes I/O and speeds up analytics.

Q65. Which of these functions can be used to merge DataFrames in Pandas?

A. df.combine()

B. df.join() [Correct]

C. df.stack()

D. df.connect()

Explanation: .join() and .merge() are the standard methods used to combine two DataFrames based on indices or common key columns.

Q66. Which Python keyword is used to inherit a class?

A. super

B. class

C. extends

D. class(SomeClass) [Correct]

Explanation: In Python, inheritance is declared by placing the parent class name in parentheses immediately after the child class name during definition.

Q68. What is the file format of Spark checkpoints?

A. .cp

B. .parquet

C. .chk

D. None (internal format) [Correct]

Explanation: Spark checkpoints save the state of an RDD/DataFrame reliably to an HDFS-compatible file

system in an internal, specialized binary format.

Q69. What Python feature supports concurrent programming?

A. lambdas

B. decorators

C. threading [Correct]

D. pip

Explanation: The threading module (along with multiprocessing and asyncio) provides concurrent execution pathways in Python programs.

Q70. What does np.array([[1,2],[3,4]]) create?

A. Series

B. Scalar

C. 2D array [Correct]

D. Dictionary

Explanation: Providing a list of lists to np.array() generates a two-dimensional NumPy array (a matrix).

Q71. Which tool is best for real-time big data processing in Python?

A. Pandas

B. Kafka [Correct]

C. Seaborn

D. SQLite

Explanation: Apache Kafka is an event-streaming platform optimized for high-throughput, real-time data ingestion and processing.

Q73. How do you check the number of nulls in a column in PySpark?

A. df.isnull('col')

B. df.count('col')

C. df.filter(df.col.isNull()).count() [Correct]

D. df.checkNull('col')

Explanation: Filtering the DataFrame using col.isNull() and then calling the count() action returns the exact number of null values in that column.

Q74. What does the iloc method in Pandas do?

A. Index by column name

B. Index by label

C. Index by integer position [Correct]

D. Group by location

Explanation: The .iloc[] indexer allows selection of rows and columns based exclusively on their integer-based positional indices.

Q75. Which data format is best suited for interoperability between systems?

A. JSON [Correct]

B. CSV

C. Parquet

D. HTML

Explanation: JSON (JavaScript Object Notation) is a lightweight, human-readable format that is universally supported across different languages and APIs.

Q76. What is the default shuffle partition number in PySpark?

A. 1

B. 10

C. 100

D. 200 [Correct in reality, but based on choices likely 100/200; assuming standard Spark configuration] (Note: Spark default is 200).

Explanation: In standard PySpark SQL configuration, `spark.sql.shuffle.partitions` defaults to 200. (The MCQ listed 100 as correct if 200 was absent).

Q78. What does df.select("name").distinct() do in PySpark?

A. Sorts names

B. Filters names

C. Returns unique names [Correct]

D. Counts rows

Explanation: .distinct() removes duplicate rows, so applied after select("name"), it outputs only the unique

values found in the name column.

Q79. What is the purpose of lambda in Python?

A. Anonymous functions [Correct]

B. Memory allocation

C. Looping

D. Import modules

Explanation: The lambda keyword creates small, anonymous, single-expression functions without the need for a formal 'def' block.

Q80. What is .agg() used for in PySpark?

A. Aggregation operations [Correct]

B. Convert to string

C. Create RDD

D. Build model

Explanation: .agg() applies aggregate mathematical functions (like sum, max, min, avg) over the entire DataFrame or grouped subsets.

Q81. What does .collect() do in PySpark?

A. Collects memory

B. Collects logs

C. Brings RDD data to the driver [Correct]

D. Stores in database

Explanation: The .collect() action retrieves all data from the distributed cluster nodes and loads it into the memory of the driver node.

Q83. What does the isinstance(obj, type) function check in Python?

A. Type conversion

B. Object creation

C. Inheritance

D. If an object is of a specific type [Correct]

Explanation: isinstance() verifies whether the specified object belongs to or inherits from the given class or

type.

Q84. Which is a valid Python set declaration?

A. set = [1,2,3]

B. set = {1,2,3} [Correct]

C. set = (1,2,3)

D. set = <1,2,3>

Explanation: Sets in Python are declared using curly braces {} containing elements separated by commas.

Q85. What is broadcast variable in PySpark?

A. For parallel computing

B. To share data across all nodes efficiently [Correct]

C. For real-time streaming

D. A type of RDD

Explanation: Broadcast variables securely cache a read-only variable on each worker node rather than sending a copy with every task, optimizing lookups.

Q86. Which of the following is NOT a real-time stream processor?

A. Apache Flink

B. Apache Kafka

C. Apache Spark Streaming

D. Hadoop MapReduce [Correct]

Explanation: Hadoop MapReduce is designed exclusively for batch processing. Flink, Kafka Streams, and Spark Streaming handle real-time data.

Q88. Which Python library is commonly used for serialization?

A. pickle [Correct]

B. shutil

C. copy

D. struct

Explanation: The 'pickle' module converts complex Python objects into a byte stream (serialization) so they can be saved to disk or sent over a network.

Q89. In PySpark, filter() is a...?

A. Transformation [Correct]

B. Action

C. Trigger

D. Aggregator

Explanation: filter() is lazily evaluated. It returns a new DataFrame/RDD without executing computation until an action (like show()) is called.

Q90. What is the output of bool([]) in Python?

A. True

B. False [Correct]

C. Error

D. None

Explanation: Empty collections (like [], {}, ()) and numeric zero evaluate to False in a boolean context in Python.

Q91. Which protocol is used to transmit serialized Python objects?

A. TCP

B. HTTP

C. RPC

D. pickle protocol [Correct]

Explanation: The pickle protocol defines the specific data format used by the pickle module for serializing and deserializing Python objects.

Q93. What does .repartition(n) do in PySpark?

A. Rebuilds dataset

B. Splits data evenly across n partitions [Correct]

C. Deletes partitions

D. Sorts dataset

Explanation: .repartition() reshuffles the underlying data across the cluster to precisely create 'n' equally sized partitions for load balancing.

Q94. Which method is used to remove duplicate rows in Pandas?

A. drop_null()

B. drop_duplicates() [Correct]

C. unique()

D. remove_dup()

Explanation: The drop_duplicates() method scans the DataFrame and safely removes any rows that contain identical values to a previous row.

Q95. What does yield do in Python?

A. Exits function

B. Creates a generator [Correct]

C. Starts thread

D. Stops loop

Explanation: The yield keyword turns a standard function into a generator, pausing execution and returning a value while maintaining state for the next iteration.

Q96. What is RDD.persist() used for?

A. Writes to file

B. Logs RDD

C. Caches RDD in memory/disk [Correct]

D. Deletes data

Explanation: .persist() is used to save an RDD to memory or disk with various storage levels, optimizing performance when the RDD is reused.

Q98. What is the default file format when saving with DataFrame.write.save()?

A. Parquet [Correct]

B. JSON

C. CSV

D. TXT

Explanation: By default, PySpark's DataFrameWriter saves output using the highly optimized Parquet columnar file format.

Q99. What is one major benefit of generators over lists in Python?

A. Faster processing

B. More readable

C. Lower memory usage [Correct]

D. Can be sorted

Explanation: Generators yield items one at a time lazily, meaning they do not store the entire sequence in memory, resulting in drastically lower memory footprint.

Q100. What is a DataFrame in PySpark?

A. Tabular, distributed data structure [Correct]

B. SQL table

C. Static array

D. Python dict

Explanation: A DataFrame is a distributed collection of data organized into named columns, functioning similarly to a table in a relational database.

Q101. What command is used to install packages in Python?

A. pip install [Correct]

B. python add

C. py install

D. install module

Explanation: 'pip' is the package installer for Python, and 'pip install <package_name>' downloads and installs the specified library.

Q103. Which command in Pandas resets the index of a DataFrame?

A. df.index_reset()

B. df.reset_index() [Correct]

C. df.clear_index()

D. df.drop_index()

Explanation: df.reset_index() moves the current index into a column and replaces the index with standard sequential integers (0, 1, 2...).

Q104. What does explode() return when applied to a column of arrays?

A. Nested lists

B. Split arrays into separate rows [Correct]

C. Sum of arrays

D. Error

Explanation: explode() flattens arrays inside a column, duplicating the other row values for each individual element inside the original array.

Q105. What does df.fillna(0) do in Pandas?

A. Removes 0s

B. Fills NA values with 0 [Correct]

C. Deletes rows

D. Creates new NA values

Explanation: The .fillna() method replaces all missing (NaN/NA) values within the DataFrame with the specified argument (in this case, 0).

Q106. What does .selectExpr() do in PySpark?

A. Selects rows only

B. Executes SQL expressions on columns [Correct]

C. Selects RDD

D. Filters schema

Explanation: .selectExpr() behaves like .select(), but parses strings as SQL expressions, allowing easy transformations inline.

Q107. Which library supports parallelism similar to Spark but in Python only?

A. Dask [Correct]

B. PyTorch

C. pandas

D. seaborn

Explanation: Dask is a flexible library for parallel computing in Python, providing multi-core and distributed execution for data science workloads.

Q108. Which method reads a parquet file in PySpark?

A. spark.read.text()

B. spark.read.csv()

C. spark.read.parquet() [Correct]

D. spark.open()

Explanation: The spark.read.parquet() function specifically loads data from files stored in the Parquet format into a DataFrame.

Q109. Which of the following is a Spark action?

A. map()

B. filter()

C. count() [Correct]

D. withColumn()

Explanation: map(), filter(), and withColumn() are lazy transformations. count() is an action that triggers the execution of the entire pipeline.

Q110. Which method is used in Pandas to combine rows with matching keys?

A. concat()

B. groupby()

C. join() [Correct]

D. append()

Explanation: While merge() is commonly used, join() also combines DataFrames horizontally by matching on indices or specified key columns.